SIMON NG

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I am a PhD Student at UCLA researching desert ecology using diverse methods, including fieldwork, remote sensing, lab work, and modeling. My background is in synthetic biology, and I have always loved living things.

EDUCATION

University of California, Los Angeles

MA/PhD Student, Geography Department *9/2025 - Present*

Cumulative GPA: 4.0

Selected Coursework: Soils & the Environment, Regional Climate & Terrestrial Surface Processes, Advanced Projects in Remote Sensing/GIS, Data Visualization, GeoStatistics

B.S., Major in **Bioengineering**, Minor in **Environmental Engineering** *9/2017 - 6/2021*

Cumulative GPA: 3.9, *magna cum laude*

Activities: Biomedical Engineering Society, Backpacking Club, Old-Time Ensemble, Community Garden

RESEARCH EXPERIENCE

UCLA Geography, *Okin Lab*
MA/PhD Student *9/2025 - Present*

- Investigate the impact of wind-driven sediment transport on growth of biological soil crusts using hyperspectral remote sensing from NEON AOP at the Jornada Experimental Range in New Mexico.

NASA Jet Propulsion Laboratory, *DEVELOP*
Remote Sensing Researcher *1/2025 - 4/2025*

- Coordinated with stakeholders to provide a remote sensing analysis of changes to mesquite bosque health in the Amargosa Basin of Death Valley using Sentinel-1 and 2, Landsat-4, 5, 8 and 9, NAIP, SMAP, and ECOSTRESS (see publication in NASA Technical Reports Server).
- Led creation of an outreach brochure for our tribal stakeholder partners.

Remote Sensing Researcher *9/2024 - 11/2024*

- Collaborated in team of 4 using hyperspectral remote sensing with a machine learning classification model to predict locations of oak trees infested with Goldspotted oak borer in the Santa Clarita Valley.
- Led development of risk maps that show areas at high risk for future oak borer infestation in Los Angeles and Ventura Counties (see publication in NASA Technical Reports Server).
- Wrote code in Python to access and process Sentinel-2 data for time series NDVI analysis.

Codexis Inc., *Protein Engineering Department*
Research Associate III *9/2021 - 4/2024*

- Evolved a 4-enzyme cascade reaction for RNAi manufacturing by expanding enzyme promiscuity while maintaining stability, enabling more sustainable synthesis of modified RNA bases (see patents).
- Co-led effort to achieve Gold-level My Green Lab certification, serving as the lab's sustainability guru.

NASA Ames Research Center, *Space Life Sciences Training Program*

Undergraduate Capstone Team Lead & Independent Researcher *9/2020 - 7/2022*

- Conceived novel research collaboration with Dr. Dino Di Carlo (UCLA) and Dr. Sergio Santa Maria (NASA Ames), then drafted a successful proposal to execute the project with a UCLA Bioengineering undergrad capstone team. Secured additional funding from NASA Ames to complete the project.
- Led research project from conception through publication, developing a preservation method to facilitate single-cell space biology research in *S. cerevisiae* (see publication in Life Microbiology).

Engineering Team Lead *6/2020 - 12/2021*

- Collaborated with 10 fellow interns to develop and execute a group project, which was a Mars-analog aerobiology experiment with *S. cerevisiae* in a custom CAD'ed payload.
- Helped set meeting agendas, lead meetings, and delegate tasks, helping to win a $50k NASA grant to fly the experiment and culminating in a successful test flight with the payload hardware.

Research Associate *6/2020 - 8/2020*

- Coded machine learning Python script to guide evolution of carbonic anhydrase for spacecraft CO_2 capture, identifying enzymes with predicted thermostability >85°C using Gaussian process regression.

UCLA Mathematics, *Research Experience for Undergraduates*

Experiment Lead *6/2019 - 7/2021*

- Saved time and money by designing and building a 3D-printed, motorized test rig to enable experiments for a bio-microfluidics x math-modeling collaboration (see publication in PhysRevE).
- Mentored high school student in wet lab techniques and CAD.

UCLA Bioengineering, *Khademhosseini Lab*

Undergraduate Researcher *1/2018 - 9/2018*

- Taught 24 lab members bioprinting techniques and provided bioprinting advice for 6 projects (see publication in Biomaterials Science).

OUTDOORS EXPERIENCE

Camp Tawonga, *Wild Department*

Backpacking and Challenge Course Leader *5/2024 - 8/2024*

- Led kids aged 7-14 on 1-2 night camping and backpacking trips in and around Yosemite National Park.
- Planned and facilitated low and high-element challenge course activities for kids aged 7-17, focusing on team building, teamwork skills, and individual confidence.

Badlands Dinosaur Museum, *Montana Field Dig*

Volunteer Crew Member *6/2023*

- Lived the dream excavating dinosaur fossils in the badlands of north-central Montana for 3 weeks.

Boy Scouts of America

Eagle Scout *6/2016 - 6/2017*

- Trained and led 24 volunteers to survey health and distribution of trees, undergrowth, and invasive species along 4 transects in a 145-acre nature preserve in Wisconsin.
- Presented biodiversity findings to nature preserve members and analyzed data in written report.

ADDITIONAL EXPERIENCE

NASA Ames Research Center, *Space Life Sciences Training Program*

Staffer *5/2021 - 8/2021*

- Supported interns and program directors with logistics work, communication, and conflict resolution.

UCLA Biomedical Engineering Society, *Design Team*

Team Lead *9/2019 - 6/2020*

- Recruited and led multi-disciplinary 5-member team to design mechanized spray bottle with integrated sensor and motor.

- Transitioned project goals from physical prototype to technical report to produce high-quality results despite COVID-19.

CAD & 3D-Printing Expert *10/2018 - 6/2019*

- Collaborated in team of 4 to design an automated pill dispenser, integrating Arduino, electrical hardware, and custom 3D-printed parts into functional prototype.

UCLA Dorms, *Housing & Hospitality / Residential Life*

Assistant Sustainability Manager *6/2018 - 9/2018*

- Automated sustainability analysis of 22 buildings' utilities and $21 million of annual food expenditures with custom Excel VBA macros.

Sustainable Living Community Development Intern *9/2017 - 6/2018*

- Managed $1300 budget and co-programmed with Resident Assistants for over 25 sustainability-related events both on and off the UCLA campus for 90 freshman residents.

PUBLICATIONS & PATENTS

1. **Ng, S.**; Williamson, C.; van Zee, M.; Di Carlo, D.; Santa Maria, S.R. Enabling Clonal Analyses of Yeast in Outer Space by Encapsulation and Desiccation in Hollow Microparticles. *Life* **2022**, *12*, 1168
2. Shen, G.; Gallegos, A.; **Ng, S.**; Blatchford, P. Amargosa Basin Ecological Conservation: Evaluating the Health of the Mesquite Bosque in the Amargosa Basin using Earth Observations. *NASA Technical Reports Server* **2025**
3. Elowitt, M.; Pendergraft, J.; **Ng, S.**; Lai, N. Santa Clarita Valley Ecological Conservation: Identifying Oak Woodland Infested with the Goldspotted Oak Borer in the Santa Clarita Valley Using Earth Observations. *NASA Technical Reports Server* **2024**
4. Entwistle, D.; Forget, S.; Knight, A.M.; Krawczyk, M.J.; MacAvoy, M.G.; **Ng, S**.; Porter, N.; Provencher, P.; Shoubber, A.; Teadt, L.Q.; et al. Enzymatic synthesis of NTP and NQP. Patent US20240209406A1 **2024***, pending*
5. Forget, S.; MacAvoy, M.G.; **Ng, S**.; Porter, N.; Teadt, L.Q.; Vroom, J. Engineered adenosine kinase variants. Patent US20250122486A1 **2025***, pending*
6. Forget, S.; MacAvoy, M.G.; **Ng, S**.; Porter, N.; Vroom, J. Engineered acetate kinase variants. Patent US20250122487A1 **2025***, pending*
7. Forget, S.; Knight, A.M.; MacAvoy, M.G.; **Ng, S**.; Porter, N.; Vroom, J. Engineered adenylate kinase variants. Patent US20250122488A1 **2025***, pending*
8. Knight, A.M.; **Ng, S**. Engineered RNase inhibitor variants. Patent US2023085570 **2024**
9. Entwistle, D.; Fell, J.; Jenne, S.; Knight, A.M.; **Ng, S**.; Reeves, R.D.; Seibel, Z.M.; Vroom, J. P450-BM3 variants with improved activity. Patent US20240318152A1 **2024***, pending*

10. Harandi, B.; **Ng, S**.; Liddell, L.C.; Gentry, D.M.; Santa Maria, S.R. Fluidic-Based Instruments for Space Biology Research in CubeSats. *Front. Sp. Technol.* **2022**, *3*, 853980
11. Du, R.S.; Liu, L.; **Ng, S.**; Sambandam, S.; Hernandez Adame, B.; Perez, H.; Ha, K.; Falcon, C.; De Rutte, J.; Di Carlo, D.; et al. Statistical Energy Minimization Theory for Systems of Drop-Carrier Particles. *Phys. Rev. E* **2021**, *104*, 015109
12. Murphy, R.; Kordbacheh, S.; Skoulas, D.; **Ng, S.**; Suthiwanich, K.; Kasko, A.M.; Cryan, S.A.; Fitzgerald-Hughes, D.; Khademhosseini, A.; Sheikhi, A.; et al. Three-Dimensionally Printable Shear-Thinning Triblock Copolypeptide Hydrogels with Antimicrobial Potency. *Biomater. Sci.* **2021**, *9*, 5144–5149

PRESENTATIONS

1. "Identifying Oak Woodland Infested with the Goldspotted Oak Borer in the Santa Clarita Valley Using Earth Observations", LA County GIS Day, **poster**. Los Angeles, CA, 2024.
2. "Codexis' Enzyme Engineering Platform", Codexis Bioprocess Development group meeting, **presentation**. Redwood City, CA, 2023.
3. "Radiation Biology Research at an Elevated Altitude through Dosimetry (RadBREAD): Designing a flexible payload platform for Mars-analog space biology research", American Society for Gravitational and Space Research, **presentation**. Baltimore, MD, 2021.
4. "Enabling single cell research in hollow hydrogel microparticles via desiccation", American Society for Gravitational and Space Research, **poster**. Baltimore, MD, 2021.
5. "Enzyme engineering for spacecraft CO2 removal using machine learning", American Association for the Advancement of Science, **e-poster presentation**. Virtual, Planet Earth, 2021.
6. "Engineering thermostable carbonic anhydrases for spacecraft CO_2 removal using machine learning", American Society for Gravitational and Space Research, **e-lightning talk**. Virtual, Planet Earth, 2020
7. "Fox Hill Nature Preserve Ecological Survey Results", Annual Meeting of the Cedar Lakes Conservation Foundation, **presentation**. West Bend, WI, 2016

CERTIFICATIONS

Wilderness First Responder	*Expires 3/2026*
CPR & AED	*Expires 3/2026*
American Red Cross Lifeguard	*Expires 5/2026*
My Green Lab Ambassador	

AWARDS & HONORS

UCLA Geography Department Award, $36,000	***9/2025 - 8/2026***
Codexis SPOT Award	*2022 & 2023*
1st place undergrad Biochemistry e-poster, American Association for the Advancement of Science	*2021*
2nd place student lightning talk, American Society for Gravitational and Space Research	*2020*
Muriel K. and Robert B. Allan Fund award, $20,000	*2020-2021*
GE-Reagan Foundation Scholar, $40,000	*2017-2021*

UCLA Coca Cola Achievement, $40,000	*2017-2021*
Herb Kohl Award, $10,000	*2017-2021*
National Merit Scholarship, $2,500	*2017*
UCLA Student Sustainability Leader	*2018*
Dean's Honor List	*2017-2021*
Eagle Scout, Boy Scouts of America	*2017*